Pergamon

Pattern Recognition, Vol. 29, No. 11, pp. 1863–1876, 1996
Copyright © 1996 Pattern Recognition Society. Published by Elsevier Science Ltd
Printed in Great Britain.
0031-3203/96 $15.00 + .00

PII: S0031-3203(96)00030-1

RECOGNITION OF HUMAN FRONT FACES USING KNOWLEDGE-BASED FEATURE EXTRACTION AND NEURO-FUZZY ALGORITHM

S. Y. LEE, Y. K. HAM and R.-H. PARK*
Department of Electronic Engineering, Sogang University, C.P.O. Box 1142, Seoul 100-611, Korea

(*Received* 23 *March* 1995; *in revised form* 17 *October* 1995; *received for publication* 28 *February* 1996)

Abstract—A recognition method of human front faces using knowledge-based feature extraction and a neuro-fuzzy algorithm is proposed. In the preprocessing step we extract the face part from the homogeneous background by tracking face boundaries, where we assume that the face part is located in the center of a captured image. Then, based on *a priori* knowledge of human faces, we extract five normalized features. In the recognition step we propose a neuro-fuzzy algorithm that employs a trapezoidal fuzzy membership function and modified error backpropagation (EBP) algorithm. The former absorbs variation of feature values and the latter shows good learning efficiency.

Computer simulation results with 80 test images of 20 persons show that the proposed neuro-fuzzy method yields higher recognition rate than the conventional ones. Copyright © 1996 Pattern Recognition Society. Published by Elsevier Science Ltd.

Face recognition Neuro-fuzzy Features Membership function Backpropagation

1. INTRODUCTION

Human face recognition is required in our daily lives and a number of implementation approaches by computers and hardware have been proposed.[1–13] It can be applied to video phone, teleconference, credit-card verification, criminal identification, security system, and so on. We can easily distinguish persons based on our experience and knowledge of their facial features, however, it is not easy to implement our experience or knowledge to a machine for human face recognition.

Many recognition methods using partial knowledge have been proposed. They include approaches based on the human face profile[2–6] or human front face[7–12] and the hybrid method[13] of these two approaches. With recognition methods employing the human face profile, feature extraction is easy. However, to increase the recognition rate we must accurately extract the face part from the background and the face should not be rotated much for reliable extraction of the face part. Using the front face, a number of approaches has been proposed such as the region matching method using second-order statistics in a binary image,[10] template matching method[11] and a feature-based method using relationships among components of the human face.[7] In a region matching method region extraction is sensitive to edge variation by the illumination. A template matching method is dependent on the captured image size. To solve these problems, we propose a neuro-fuzzy algorithm utilizing the positional relationship among components of the human face.

In human face recognition the face profile, the shape and size of eyes, nose, mouth and their relationships have been commonly used as features. In conventional face recognition methods many approaches employed features related to contour boundaries of the side of a face.[1] With correctly extracted features we can easily recognize human faces. However, in separating the face part from the background, the face shape may be distorted by the shadow, hair, noise or rotation of a face. Thus, we propose a human face recognition algorithm based on five efficient features extracted from a front face image. These features are normalized and then fuzzified, thus they become less sensitive to variation of feature values of the same person.

After extracting facial components such as eyes, nose and mouth we recognize human front faces using a proposed neuro-fuzzy algorithm, in which a fuzzy membership function absorbs the variation of normalized feature values of the same person, whereas neural networks have good learning efficiency. This paper is structured as follows. In Section 2 we describe the preprocessing step for feature extraction and in Section 3 we present knowledge-based feature extraction. In Section 4 we review conventional recognition algorithms and propose a neuro-fuzzy algorithm. In Section 5 we present recognition results and discuss experimental results. Finally, we give conclusions in Section 6.

* Author to whom correspondence should be addressed.

z_0

1	2	1
0	0	0
−1	−2	−1

z_1

2	1	0
1	0	−1
0	−1	−2

z_2

1	0	−1
2	0	−2
1	0	−1

z_3

0	−1	−2
1	0	−1
2	1	0

Fig. 1. Sobel edge operators.

2. PREPROCESSING

We use 3×3 median filtering to reduce spot noise in an input image. For edge extraction, four-directional Sobel edge operators with locally adaptive threshold (LAT)[14] are employed. For accurate extraction of features the face part is separated from the background and shoulders by morphological operations.

2.1. *Edge extraction*

Spot noise is successfully removed using 3×3 median filtering. After filtering noise, we use four-directional Sobel edge operators along with LAT. LAT is expressed as:

$$\mathrm{LAT}(x,y) = \frac{\mathrm{Max}\{|z_k(x,y)|, \quad k=0,1,2,3\}}{g(x,y)}, \tag{1}$$

where $g(x,y)$ denotes the low-pass filtered version of an input image and $z_k(x,y)$ represents the output of the k-th directional Sobel edge operator, as shown in Fig. 1. The low-pass filter is specified by a 3×3 mask:

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}. \tag{2}$$

Then we obtain a binary edge map based on the decision rule:

$$\mathrm{pixel}(x,y) = \begin{cases} 1 & \text{(edge point)}, \quad \text{for } \mathrm{LAT}(x,y) > 1 \\ 0 & \text{(nonedge point)}, \quad \text{otherwise.} \end{cases} \tag{3}$$

In a binary edge map black (white) points denote edge (nonedge) points with a binary value of 1 (0). Generally, the eye can perceive intensity changes more easily in dark regions than in light regions, so we generate a binary edge map using LAT.

2.2 *Separation of the face part from the background*

For separation of the face part from the background, Kelly[8] used a planning method after decimating an original image by a factor of 64 and Craw *et al.*[9] employed an 8×8 template of the coarse facial outline. Kato *et al.*[15] used the longest isodensity line of an edge image after requantizing an input image to eight gray levels. Harasaki *et al.*[16] used stereo image focusing.

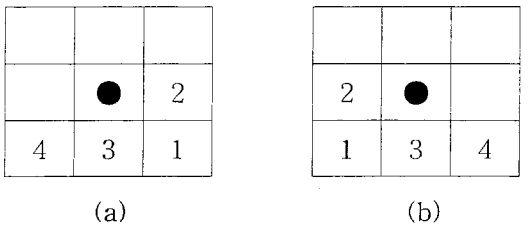


Fig. 2. Windows for tracking face boundaries: (a) bottom-left window; (b) bottom-right window.

We apply a knowledge-based facial boundary tracking method[7] to a binary edge map. Since edges need to be linked for effective separation of the face part from the background, we use morphological dilation with a 3×3 window. Morphological dilation[17] is defined by:

$$A \oplus B = \bigcup_{b \in B} (A + b), \tag{4}$$

where A represents a 3×3 window and B denotes a binary facial image with its element b representing an edge point.

The detailed separation procedure is described as follows. First, check the binary value at a starting point (0, 100), where (0, 0) at the top-left point represents an origin of a 256×256 input image. If the starting point is black, it is determined as the top-left point. Otherwise, trace along the x direction and the first black pixel encountered is determined as the top-left point. From the top-left point we trace left boundaries of a face. Similarly, check the binary value at a starting point (255, 100). If that point is black, it is determined as the top-right point. Otherwise, trace along the $-x$ direction and the first black pixel detected is determined as the top-right point. From the top-right point we trace right boundaries of a face. In finding the top-left and -right points of the face, the y-axis value is set to 100 with the assumption that the face is located in the center of an image. Windows for tracking left and right boundaries are shown in Fig. 2(a) and (b), respectively, where numbers denote priority in boundary tracking, 1 representing the highest priority. The separated binary face images in the preprocessing step are shown in Fig. 3(b) and (e) for input face images of Fig. 3(a) and (d), respectively.

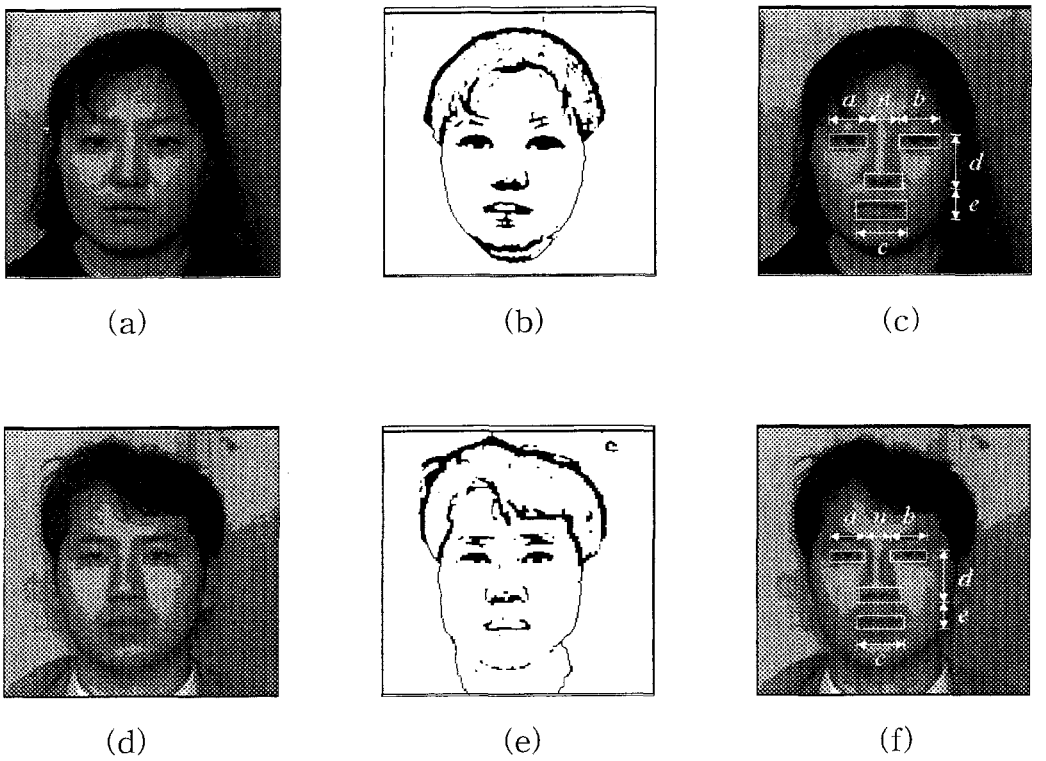


Fig. 3. Facial feature extraction: (a) face image 1; (b) binary face image 1 separated from the background and shoulders; (c) feature extraction results for the face image 1; (d) face image 2; (e) binary face image 2 separated from the background and shoulders; (f) feature extraction results for the face image 2.

3. KNOWLEDGE-BASED FEATURE EXTRACTION

In conventional methods the commonly used features for face recognition include widths of eyes, nose and mouth, distances between a pair of facial components and distances from nose and mouth to sides of a face. However, if sides of a face are used in determining feature values, it is dificult to obtain reliable feature values since the face part is sometimes incorrectly separated because of shadow, noise or hair. Thus, we use five normalized features which are not related to facial boundaries. These features must be extracted accurately, thus we assume the restricted conditions as follows. First, an input image shows a front face located in the center of an image. Second, eyes are open and mouth is closed. Third, the rotated angle of an obtained image must be small, e.g. less than 5°. Finally, the face part should not be covered by shadow, glasses, hair, beard, mustache, and so on. Under these assumptions, we propose a feature extraction method using generic knowledge of faces. In addition, because tilt and rotation of an input image decrease the recognition rate, we compensate for tilt and rotation with an elliptic face model.[7]

3.1. *Knowledge representation for facial feature extraction*

We propose a knowledge-based feature extraction method for recognition of front faces. With parameters shown in Fig. 4, the general knowledge-based for feature extraction can be represented. In Fig. 4 the origin is defined at the top-left point, x_1 and x_3 denote

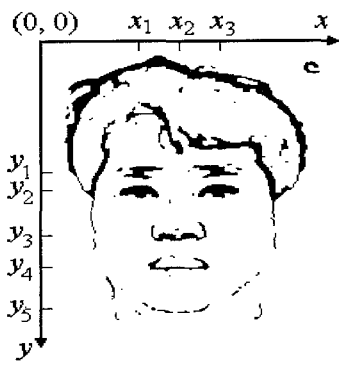


Fig. 4. Parameters for knowledge representation of a face image.

row positions of two eyes and x_2 represents the row position of nose and mouth y_1, y_2, y_3, y_4 and y_5 denote column positions of the brow, eyes, nose, mouth and chin, respectively. Knowledge used for facial feature extraction is represented by:

Knowledge 1. The largest horizontal projection value determines the positions of eyes or brows.
Knowledge 2. $\overline{y_2y_3} > \overline{y_3y_4}$.
Knowledge 3. $\overline{x_1x_2} \cong \overline{x_2x_3}$.
Knowledge 4. $x_1 < x_2 < x_3$.
Knowledge 5. $y_1 < y_2 < y_3 < y_4 < y_5$.
Knowledge 6. $\overline{x_1x_2}:\overline{y_2y_4} \cong 1:1.5$.

We can explain above knowledge as follows. Knowledge 1 represents that we are able to determine the eye's location by considering horizontal projection values. Horizontal (vertical) projection values are defined by the sums of binary values along the horizontal (vertical) direction, thus the horizontal (vertical) projection value at the specific $y(x)$ value is equivalent to the number of edge points at a specific row (column). We validate the detected eye's location using Knowledge 2–4. Knowledge 2 signifies that the vertical distance from eye to nose is larger than that from nose to mouth, whereas Knowledge 3 means that the face is symmetric with the nose in the center of an image. Knowledge 4 explains the relative positional information of the left eye, nose and right eye. Similarly, Knowledge 5 represents the relative positional information of the brow, eye, nose, mouth and chin. Knowledge 6 means that the ratio of height to width of the specific facial region is *ca* 1.5.

3.2. *Feature extraction*

Using knowledge described in the previous subsection, we present the proposed facial feature extraction algorithm as follows. First, we calculate horizontal projections at each row over the whole image and determine the y position of eyes based on the fact that the horizontal projection value of the eye part is the largest. Then, below the eye position we extract two largest horizontal projection values, which correspond to nose and mouth. If the vertical distance between the detected nose and eye is smaller than that between the detected nose and mouth, the processing is restarted after deleting projection values near the detected eye position. In that case, we assume that the eye position is detected incorrectly because of hairs, brow or nose. The x positions of the two eyes are determined by finding two regions having large vertical projection values computed over the limited interval of ± 8 pixels from the y-axis value of the detected eye position. Similarly, if a center point of nose (mouth) lies between two inner end points of right and left eyes, vertical projections are computed over the limited range of ± 8 pixels from the y-axis value of the detected nose (mouth) position and then the x-axis value giving the longest vertical projection is determined as the x position of the nose (mouth). If the center point of the detected nose does not lie between the middle of the forehead, then restart feature extraction from the first step after deleting the horizontal projection values of eyes, because the detected part may be brow or hair part. Finally, we determine accurately the y-axis value of each part by selecting the largest horizontal projection value computed over the limited interval of ± 10 pixels from the y-axis value of the detected facial component.

Next, each facial component is defined by a minimum bounding rectangle (MBR). Results of facial feature extraction ("eye", "nose" and "mouth" rectangles) are shown in Fig. 3(c) and (f), where widths (a and b) of two eyes, width (c) of mouth and vertical distance (d) from the upper side of the "eye" rectangle of the lower side of the "nose" rectangle and vertical distance (e) from the lower side of the "nose" rectangle to the lower side of the "mouth" rectangle are superimposed. These feature values are normalized by the distance (n) between two inner sides of right and left "eye" rectangles to compensate for their dependence on image size.

3.3. *Compensation for tilt and rotation of a face image*

Feature values detected in a 2D image vary depending on the viewing angle of a face. Therefore, for reliable extraction of feature values we must compensate for tilt or rotation of a face. To increase the recognition rate, we use a scheme compensating for tilt and rotation of a face.[7] We first assume that the face can be modeled as an ellipse, as shown in Fig. 5(a). In Fig. 5(a), although a modeled facial image is rotated around the y-axis, the distance $\overline{AB}$ varies slightly. However, it is difficult to find accurately points A and B because of possible occlusion by hair, shadow or noise. So, we replace points A and B by points i and j, respectively, as reference points for compensation of face rotation. Figure 5(b) shows the slice view of the mouth, from which we can express M_a and M_b as:

$$M_a = r - r\cos\theta_a$$
$$M_b = r - r\cos\theta_b, \tag{5}$$

where M_a (M_b) represents the horizontal distance between points of the leftmost (rightmost) face and a mouth in a 2D image, $2r$ denotes a face width in a 2D image. Since M_a, M_b and r are known, we can calculate θ_a and θ_b as:

$$\theta_a = \cos^{-1}\left(1 - \frac{M_a}{r}\right)$$
$$\theta_b = \cos^{-1}\left(1 - \frac{M_b}{r}\right). \tag{6}$$

In Fig. 5(b), with the assumption that θ_o is fixed, we can also compute the rotation angle θ_{rot} about the y axis as:

$$\theta_{\text{rot}} = \frac{\theta_b - \theta_a}{2}. \tag{7}$$

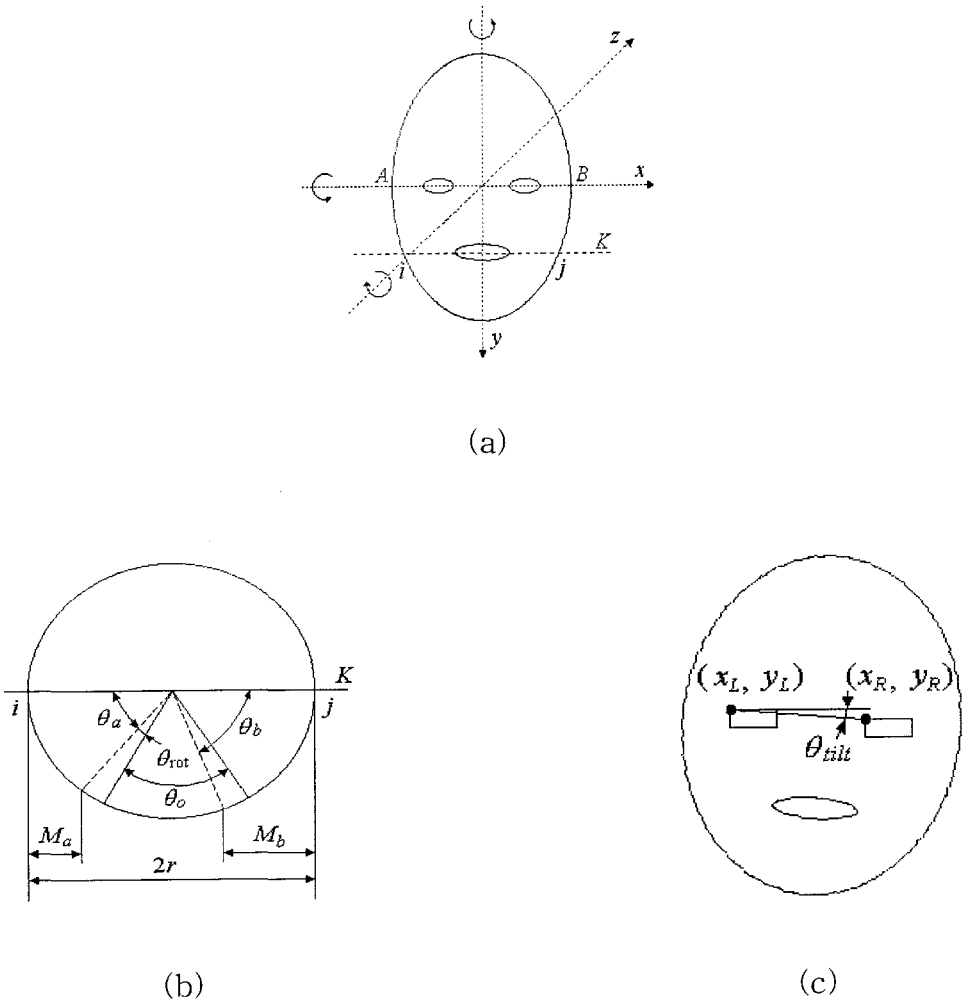


Fig.5. Face image modeling: (a) elliptic face model; (b) slice view at K; (c) rotated face image about the z axis.

Using equation (7), we compensate for the rotation angle about the y axis.

Figure 5(c) shows a model for a rotated face, in which the tilt angle θ_{tilt} about the z axis is defined by:

$$\theta_{\text{tilt}} = \tan^{-1}\frac{y_R - y_L}{x_r - x_L}, \tag{8}$$

where (x_R, y_R) and (x_L, y_L) represent the top-left points of the detected left and right "eye" rectangles, respectively. Using equation (8), we compensate for the tilt angle about the z axis.

4. FACE RECOGNITION

In this section we present conventional face recognition methods. Also we propose a neuro-fuzzy face recognition algorithm using five normalized features described previously.

4.1. *Face recognition based on the Euclidean distance*

In conventional recognition methods, the Euclidean distance has been commonly used as a distance measure between two vectors.[3–5] The Euclidean distance d is defined by:

$$d = \left(\sum_{i=1}^{L} (X_i - R_i|^2 \right)^{1/2}, \tag{9}$$

where X_i and R_i represent the ith components of the input and reference feature vectors, respectively, and I denotes the total number of features. Or the normalized Euclidean distance d_N is defined by:

$$d_N = \left(\sum_{i=1}^{I} \frac{|X_i - R_i|^2}{\sigma_i^2} \right)^{1/2}, \tag{10}$$

where the normalization constant σ_i represents the standard deviation of the ith feature. In our study, two

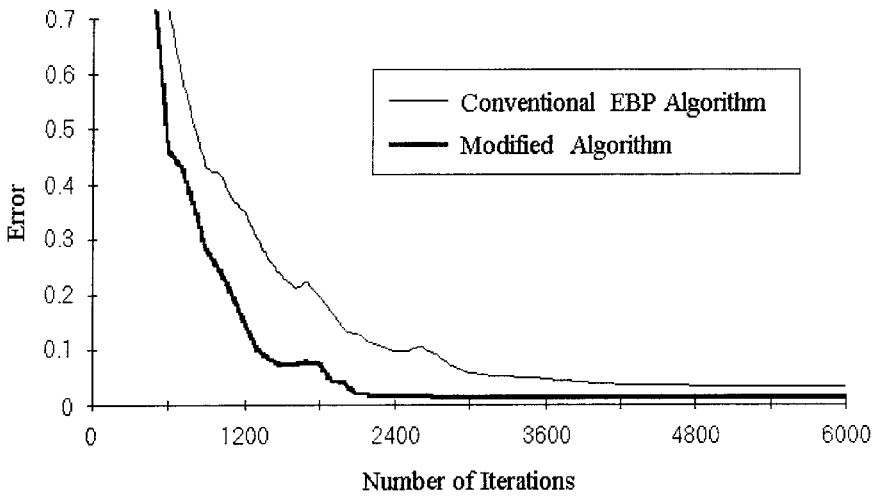


Fig. 6. Convergence characteristics of the error backpropagation algorithm.

normalization schemes are considered:

$$\sigma_i = \left(\frac{1}{N} \sum_{n=1}^{N} \left(\frac{1}{K} \sum_{k=1}^{K} T_{ink} - \frac{1}{N} \sum_{n=1}^{N} \left(\frac{1}{K} \sum_{k=1}^{K} T_{ink} \right) \right)^2 \right)^{1/2}, \tag{11}$$

$$\sigma_i = \left(\frac{1}{N} \sum_{n=1}^{N} \left(\frac{1}{K} \sum_{k=1}^{K} \left(T_{ink} - \frac{1}{K} \sum_{k=1}^{K} T_{ink} \right)^2 \right) \right)^{1/2}, \tag{12}$$

where K represents the number of images for the same person in training, N denotes the total number of persons in experiments and T_{ink} represents the ith component of the feature vector in training the kth image of the nth person. In our study, we use 80 test images with four images for each person: test images consist of four sets of images, each set consisting of 20 persons. In experiments, 20 persons are to be recognized and three images are used to construct a database for the same person, i.e. $N = 20$ and $K = 3$. The final recognition rate is obtained with 80 test images. Note that, as a mean of feature values, equation (11) uses the average feature value of all persons, whereas equation (12) uses the average feature value of the same person.

4.2. *Face recognition using the EBP algorithm*

In this subsection face recognition using an EBP algorithm[18] is discussed. Conventional EBP algorithms used a fixed learning rate and momentum factor, thus to reduce the learning time and to avoid local minima we must adaptively determine these parameters. We use a variable learning rate and momentum factor by iteratively updating weights, resulting in the modified EBP algorithm. In the modified EBP algorithm, the learning rate η, momentum factor α, and weight ω are updated by the following equations, respectively:

$$\Delta\eta_{jk}(m+1) = \varepsilon \left(\frac{\partial E_p}{\partial \omega_{jk}} \right)^2 + \beta \Delta\eta_{jk}(m), \tag{13}$$

$$\Delta\alpha_{jk}(m+1) = \mu \left(\frac{\partial E_p}{\partial \omega_{jk}} \right)^2 + \gamma \Delta\alpha_{jk}(m), \tag{14}$$

$$\Delta\omega_{jk}(m+1) = -\eta_{jk}(m+1) \frac{\partial E_p}{\partial \omega_{jk}} + \alpha_{jk}(m+1) \Delta\omega_{jk}(m), \tag{15}$$

where ε, β, μ and γ denote constants and m represents the iteration step. All the constant values are set to 0.3. Subscripts j and k signify the jth neuron of the input (hidden) layer and kth neuron of the hidden (output) layer, respectively. E_p represents the total error function at the pth layer. In Fig. 6, we show the convergence characteristics of the conventional and modified EBP algorithms. In learning we use five input neurons, 22 hidden neurons in the hidden layer and 20 output neurons for 20 persons. Parameter values are selected experimentally: in the conventional algorithm, the learning rate and momentum factor are set to 0.9 and 0.2, respectively, whereas in the modified algorithm, 0.4 and 0.2, respectively. Figure 6 shows that the modified EBP algorithm converges much faster than the conventional one.

4.3. *Proposed face recognition using a trapezoidal fuzzy membership function*

Since the fuzzy concept was proposed by Zadeh in 1965, many applications to various areas have been presented.[19,20] There are several types of membership functions in representing fuzzy phenomena. We simulate the conventional face recognition algorithm[12]

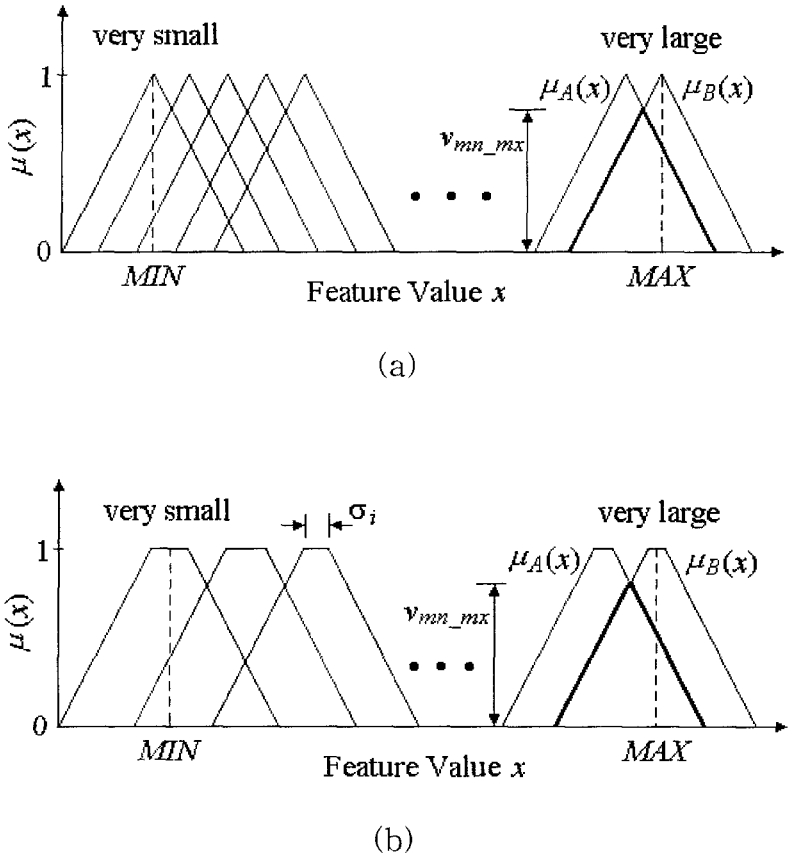


Fig. 7. Linguistic representation of feature values: (a) triangular membership function; (b) trapezoidal membership function.

using triangular membership functions and the proposed face recognition algorithm using trapezoidal membership functions.[21]

In the face recognition method using triangular membership functions, we represent five features by means of linguistic variables specified by a triangular membership function as shown in Fig. 7(a), where MIN and MAX represent the minimum and maximum feature values obtained in constructing a database, respectively. The detailed scheme is explained as follows. We first convert feature values into linguistic variables. The number of linguistic variables N_l depends on the total number of persons N and the min–max value v_{mn-mx} between neighboring linguistic variables, which is defined by:

$$v_{mn-mx} = \max(\min[\mu_A(x), \mu_B(x)]), \tag{16}$$

where $\mu_A(x)$ and $\mu_B(x)$ denote membership functions of fuzzy sets A and B, respectively. v_{mn-mx} is illustrated in Fig. 7(a) and (b) for triangular and trapezoidal membership functions, respectively. With N_l larger than N, if v_{mn-mx} is very small, i.e. the membership function becomes sharp, then the recognition rate is small. So we should select N_l properly by considering the shape and type of the membership function.

In the recognition step, we calculate the min–max value between the membership function of a given feature of a specific person and that obtained from an input image, then we use the average of v_{mn-mx} values of five features as a similarity measure O_n, which is defined by:

$$O_n = \frac{1}{I}\sum_{i=1}^{I} \max(\min[\mu_A^n(x_i), \mu_B(x_i)]), \quad 1 \le n \le N, \tag{17}$$

where $\mu_A^n(x_i)$ and $\mu_B(x_i)$ represent the membership functions of the nth person in a database and the one computed from an input image, respectively, for the ith feature. Note that we use five features and that there are 20 persons in a database: $I = 5$ and $N = 20$. The input image is recognized as a person giving the largest similarity measure O_n.

If a triangular membership function is used to represent a linguistic variable, we cannot properly consider variation of feature values for the same person. Thus, we propose a face recognition algorithm using a trapezoidal fuzzy membership function. The trapezoidal membership function, as shown in Fig. 7(b), locates at the average value of features of the same person, and

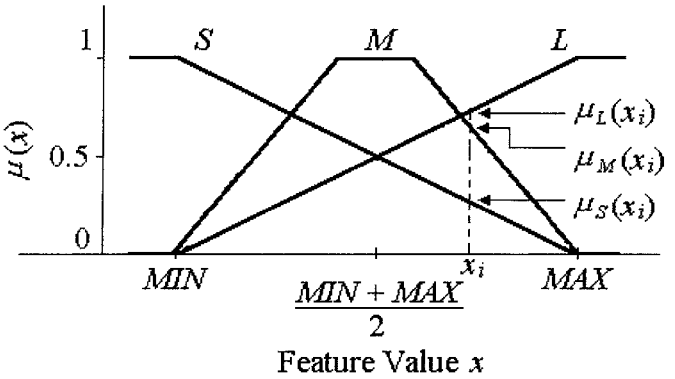


Fig. 8. Trapezoidal membership functions.

has a maximum value of 1 over the limited range which is specified by the standard deviation of the feature value. To generate a linguistic variable we first compute the average and standard deviation of the feature values of the same person. Then we uniformly divide the interval between MIN and MAX into several subintervals, where MIN and MAX represents the minimum and maximum of average values of the specific feature, respectively. The membership function of each person is centered at the average value of the features of the same person. Variation of feature values for the same person is allowed by employing the trapezoidal membership function, i.e. the width at the top of the trapezoidal membership function is set to σ_i, where σ_i denotes the standard deviation of the ith feature value, computed by equations (11) or (12). These membership functions for 20 persons are stored in a database. Final recognition is performed based on the similarity measure O_n as previously described.

4.4. *Proposed face recognition using the neuro-fuzzy algorithm*

To increase the recognition rate, we employ a neuro-fuzzy algorithm that combines neural networks and the fuzzy concept. Neural networks have learning capability and the fuzzy concept can absorb variability in feature values. The fuzzy concept can be combined with neural networks in various ways. For example, the fuzzy concept was used to change the learning rate or momentum factor.[22] This algorithm was efficient to overcome the local minima problem with reduced computational complexity. We apply this scheme to face recognition, however, it does not increase the recognition rate. Thus, we apply the fuzzy concept simply in converting feature values into fuzzified data, which are inputs to the EBP algorithm. In the proposed neuro-fuzzy algorithm we use the fuzzy data as inputs to neural networks. Sometimes, variation of feature values is large, then it is difficult to recognize faces correctly based on these feature values. To solve this problem, we first convert each facial feature value into three fuzzy data,[23] then learning is performed with these $3I$ fuzzy data using the EBP algorithm. Finally, we recognize faces using the EBP algorithm. To reduce the computational complexity and to avoid the local minima problem, we also use the modified EBP algorithm described in the previous subsection. From now on, the EBP algorithm refers to the modified EBP algorithm.

To convert five normalized features into 15 fuzzy data, we determine the MAX and MIN values which are the maximum and minimum feature values for 20 persons, respectively. As shown in Fig. 8 we generate three membership functions denoted by "S" (small), "M" (medium) and "L" (large). Note that these membership functions are specified by MIN and MAX, as shown in Fig. 8. Then we compute three fuzzy data for each feature value and use these data as the input data to neural networks. In Fig. 8 $\mu_S(x_i)$, $\mu_M(x_i)$ and $\mu_L(x_i)$ are three fuzzy data of an input feature value x_i, corresponding to linguistic variables of "S", "M" and "L", respectively. Construction of a database by learning with these 15 fuzzy data is the same as performed in the fuzzy algorithm using a trapezoidal membership function, which is explained in the previous subsection. Note that for input data greater (smaller) than MAX (MIN) we clip the membership value to 1 (0).

The proposed neuro-fuzzy algorithm yields higher recognition rates than the conventional algorithms: Euclidean distance method, EBP algorithm and fuzzy algorithms. If the deviation between the data in a database and test data is large, i.e. tilt or rotation of a face is large, the recognition rate by any method is low. Therefore, more efficient method compensating for large tilt and rotation is required.

5. EXPERIMENTAL RESULTS AND DISCUSSIONS

We use 80 front face images consisting of four set of images. Each set of images includes 20 persons: 14 men and six women. For each person four 256 × 256 images

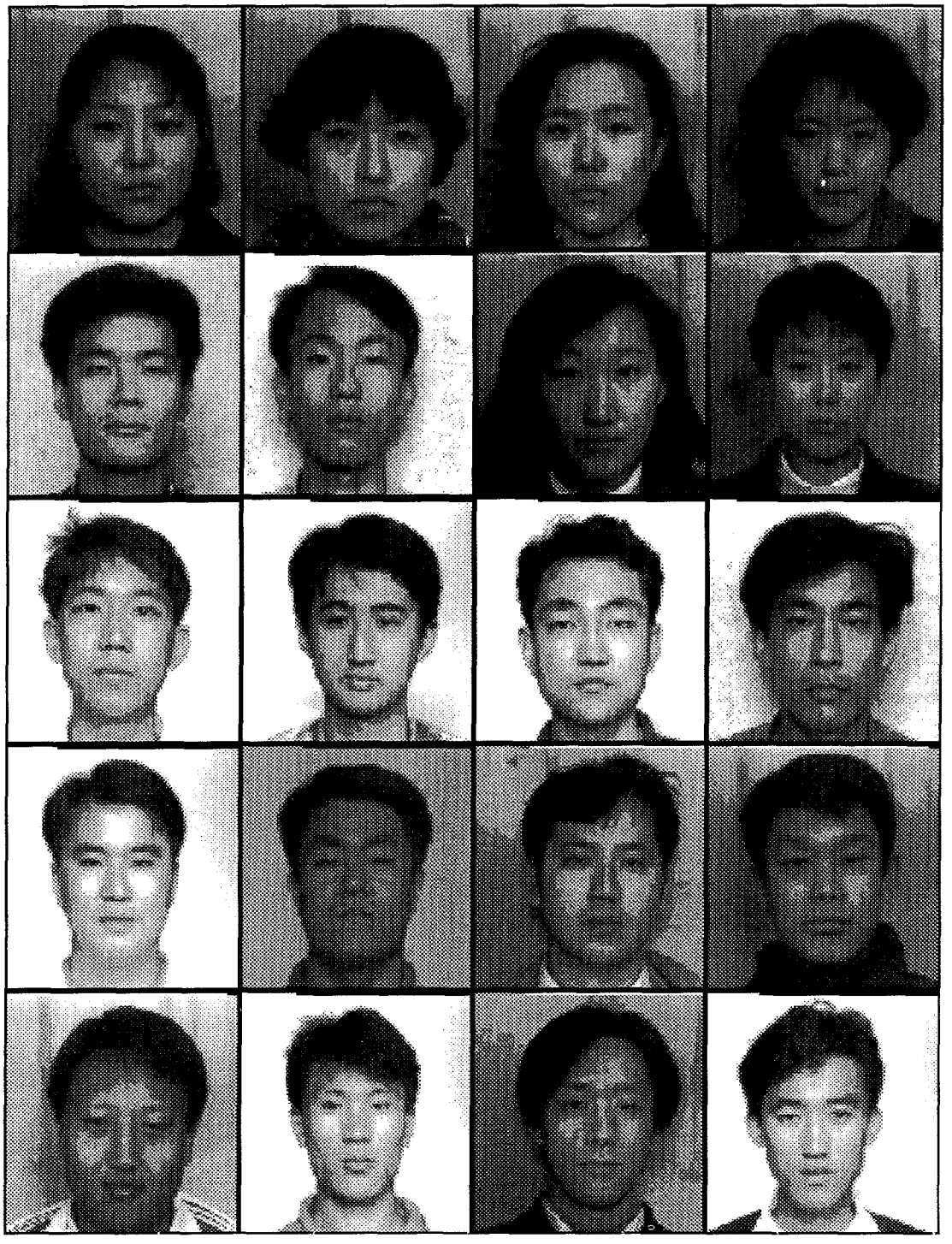
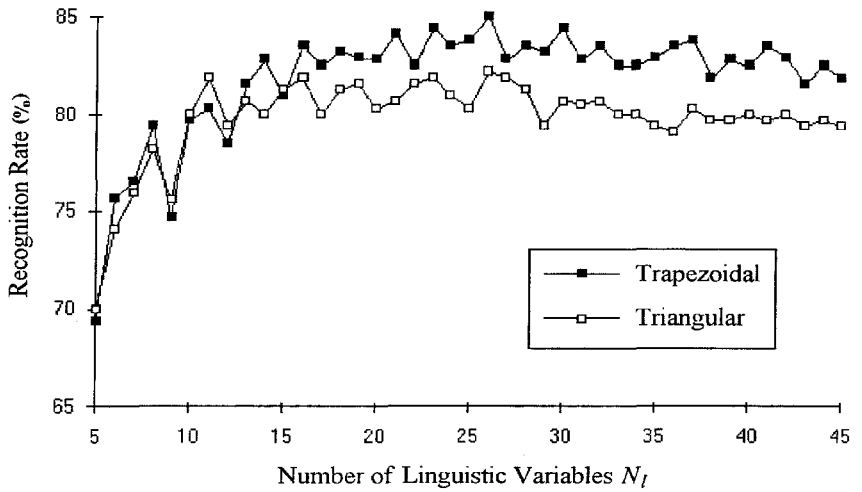


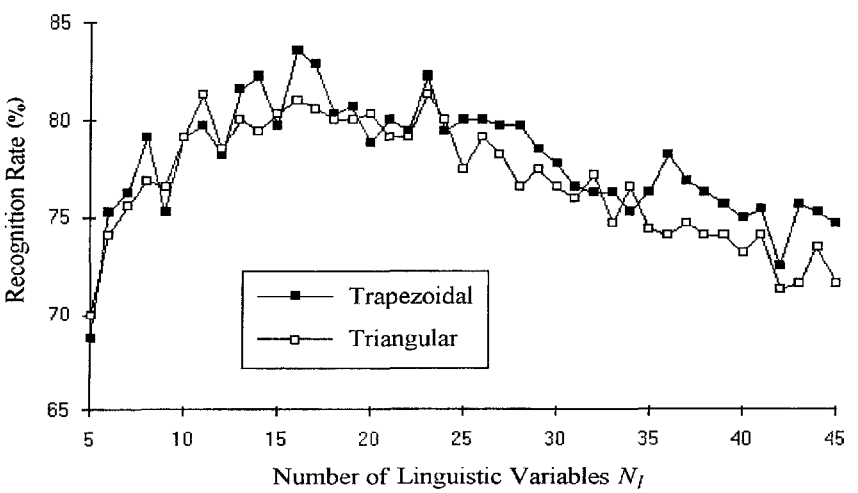
Fig. 9. A set of test images used in experiments.

Table 1. Recognition rate by the Euclidean distance methods using different normalization schemes

Normalization method	N	Recognition rate (%) *db*1	*db*2	*db*3	*db*4	Average
Not used	10	90.0	85.0	87.5	95.0	89.4
	15	90.0	85.0	88.3	86.7	87.5
	20	80.0	77.5	85.0	81.3	81.0
Using equation (11)	10	90.0	85.0	90.0	95.0	90.0
	15	88.3	83.3	90.0	88.3	87.5
	20	81.3	78.8	87.5	81.3	82.2
Using equation (12)	10	90.0	87.5	87.5	95.0	90.0
	15	88.3	86.7	90.0	88.3	88.3
	20	82.5	78.8	86.3	81.3	82.2



(a)



(b)

Fig. 10. Recognition rate by the fuzzy algorithm as a function of N_l for fixed υ_{mn-mx}: (a) $\upsilon_{mn-mx} = 0.9$; (b) $\upsilon_{mn-mx} = 0.5$.

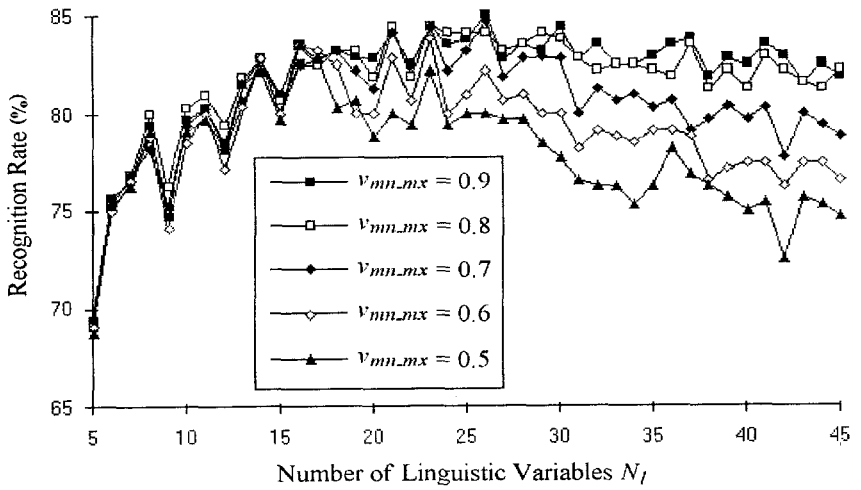


Fig. 11. Recognition rate by the proposed fuzzy algorithm with a trapezoidal membership function as a function of N_l for various v_{mn-mx}.

are obtained with different illumination conditions, each image uniformly quantized to eight bits. Training and test image sets are not disjoint. We use three sets of images to construct a database. The test is made on the entire set of images consisting of 80 facial images. One set of face images of 20 persons used in experiments is shown in Fig. 9.

For performance comparison with the proposed neuro-fuzzy algorithm, we simulate conventional algorithms such as the Euclidean distance method, EBP algorithm and fuzzy methods with triangular and trapezoidal membership functions.

In this section we present and discuss experimental results of several recognition methods. In each experiment, *db*1, *db*2, *db*3 and *db*4 denote databases constructed with three sets of images except for the first, second, third, and fourth set of images, respectively. Note that four images are obtained for each person and the recognition rate represents the percentage of the number of correctly recognized images to the total number of images (80).

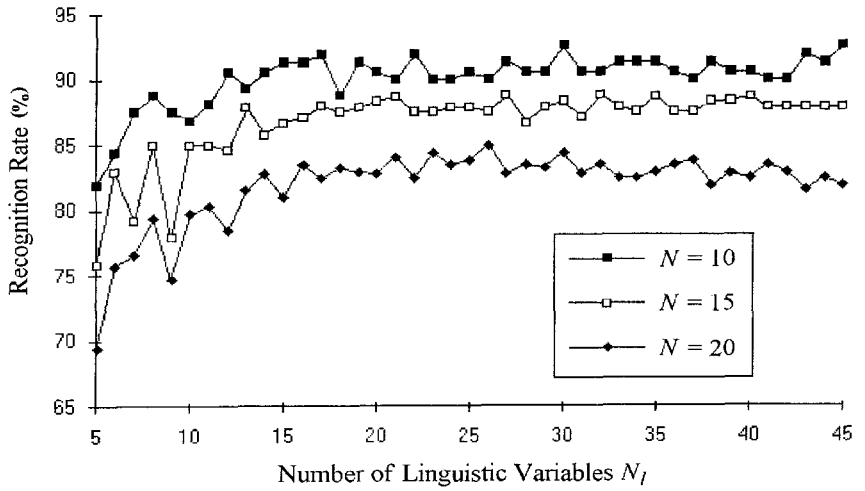
In Table 1 we show the recognition results by the Euclidean distance method as a function of N for different normalization schemes. We can observe that the recognition rate decreases as N increases. The normalized Euclidean distance method using equation (12) shows the better results, supporting the fact that appropriate normalization among different features is required for reliable recognition.

Also, we present the recognition rate of the fuzzy approach with two different membership functions: triangular and trapezoidal. Experimentally, we determine the optimal N_l and v_{mn-mx} by varying N_l from 5 to 45 with a step increment of 1 and v_{mn-mx} from 0.5 to 0.9 with a step increment of 0.1. Figure 10(a) and (b) show recognition results with v_{mn-mx} equal to 0.9 and 0.5, respectively. As shown in Fig. 10(a) and (b), in all cases the proposed fuzzy method with a trapezoidal membership function yields better results than that with a triangular membership function. In Fig. 11 we also show that recognition rate of the fuzzy algorithm with a trapezoidal membership function as a function of N_l for various v_{mn-mx}. We observe that if N_l is large and v_{mn-mx} is small, i.e. with a sharp membership function, the recognition rate becomes low. To observe variation of the recognition rate of the fuzzy algorithm with a trapezoidal membership function as a function of N, we change N: for example, $N = 10$, 15 and 20. In Fig. 12 and Tables 1–3, $N = 10$ corresponds to experizmental results with the first 10 persons in Fig. 9. Cases of $N = 15$ and 20 are similarly defined. Figure 12(a) and (b) show the recognition rate by the fuzzy algorithm with a trapezoidal membership function as a function of N for $v_{mn-mx} = 0.9$ and 0.5, respectively. We can find that if N is large, the recognition rate is low. According to the experimental results, N_l is determined by the rule of thumb:

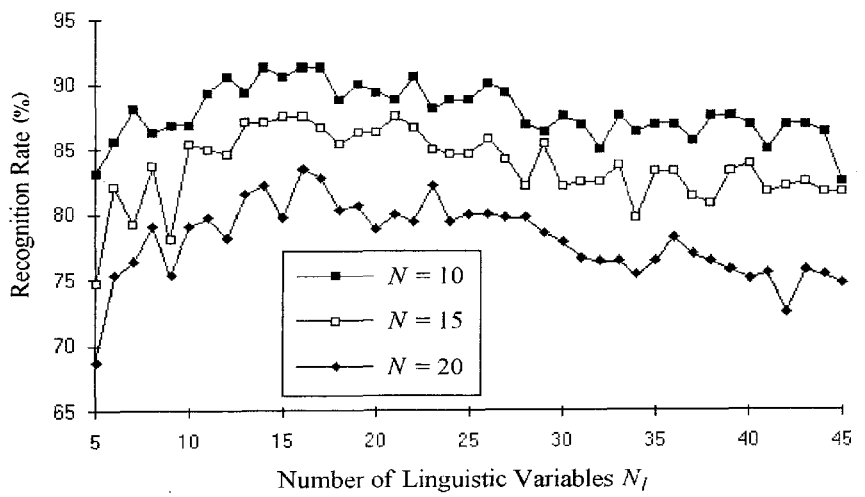
$$N_l = N + \left[\frac{1}{1.0 - v_{mn-mx}}\right], \tag{18}$$

where the symbol $[x]$ denotes the smallest integer larger than x.

With N_l determined by equation (18), Table 2 shows the recognition rate by the proposed fuzzy algorithm using a trapezoidal membership function as a function of N and v_{mn-mx}. From Table 2 we can observe that the recognition rate is the highest when v_{mn-mx} is equal to



(a)



(b)

Fig. 12. Recognition rate by the proposed fuzzy algorithm with a trapezoidal membership function as a function of N_l for various N: (a) $v_{mn-mx} = 0.9$; (b) $v_{mn-mx} = 0.5$.

0.8 or 0.9. Thus, v_{mn-mx} is set to 0.8 in computing the final recogniton rate by the proposed fuzzy method using a trapezoidal membership function.

In the modified EBP algorithm to reduce the computational complexity and to avoid local minima problem, we adaptively vary the learning rate and momentum factor. In the input layer, there exist five nodes for five features, but 20 neurons in the output layer. Of course, the number of neurons in the output layer is set to N. The number of nodes in the hidden layer is determined experimentally: in our simulation we select the number which gives the best recognition results, i.e. 22 neurons are used. In the proposed neuro-fuzzy algorithm we use 15 fuzzy data in the input layer and in the hidden and output layers we use the same number of neurons as in the modified EBP algorithm. Table 3 shows the recognition results by the EBP algorithm and proposed neuro-fuzzy algorithm for $N = 10$, 15 and 20. From Table 3 we observe that the recognition rate for large N is lower than that for small

Table 2. Recognition rate by the proposed fuzzy algorithm using a trapezoidal membership function as a function of various parameters

N	v_{mn-mx} (N_t)	Recognition rate (%) *db*1	*db*2	*db*3	*db*4	Average
	0.9 (20)	92.5	80.0	97.5	92.5	90.6
	0.8 (15)	90.0	85.0	95.0	92.5	90.6
10	0.7 (14)	90.0	80.0	97.5	92.5	90.0
	0.6 (13)	90.0	82.5	92.5	92.5	89.4
	0.5 (12)	90.0	85.0	95.0	92.5	90.6
	0.9 (20)	90.0	80.0	90.0	91.7	87.9
	0.8 (15)	90.0	80.0	91.7	91.7	88.4
15	0.7 (14)	90.0	86.7	90.0	91.7	89.6
	0.6 (13)	85.0	83.3	88.3	90.0	86.7
	0.5 (12)	88.3	81.7	88.3	88.3	86.7
	0.9 (20)	86.3	78.8	88.8	83.8	84.4
	0.8 (15)	88.8	81.3	85.0	81.3	84.1
20	0.7 (14)	82.5	81.3	86.3	78.8	82.2
	0.6 (13)	81.3	85.0	86.3	82.5	83.8
	0.5 (12)	77.5	76.3	82.5	81.3	79.4

N and the recognition rate by the proposed neuro-fuzzy algorithm is higher than that of the EBP algorithm.

In Table 4, we list the recognition results by various methods, where v_{mn-mx} is set to 0.8 in fuzzy-based methods. Table 4 shows that the proposed neuro-fuzzy algorithm using a trapezoidal membership function yields better recognition results than the Euclidean distance method, EBP method and fuzzy-based algorithms.

6. CONCLUSIONS

We propose a face recognition method using a neuro-fuzzy algorithm that combines the EBP algorithm and the fuzzy concept. Also we propose a facial feature extraction method based on the *a priori* knowledge of human faces. In the feature extraction step we extract five normalized features based on their positional relationships. These feature values are fuzzified and applied to the EBP algorithm in the recognition step.

We show simulation results of the proposed neuro-fuzzy algorithm as a function of various parameters. Simulation results show that the proposed neuro-fuzzy algorithm using a trapezoidal membership function gives better recognition results than the conventional methods. Further research will focus on development of efficient learning and recognition algorithms for a large set of face images.

REFERENCES

1. A. Samal and P. A. Iyengar, Automatic recognition and analysis of human faces and facial expressions: A survey, *Pattern Recognition* **25**, 65–77 (1992).
2. G. J. Kaufman Jr and K. J. Breeding, The automatic recognition of human faces from profile silhouettes, *IEEE Trans. Syst., Man Cybernet.* **SMC-6**, 113–121 (1976).
3. L. D. Harmon, S. C. Kuo, P. F. Ramig and U. Raudkivi, Identification of human face profiles by computer, *Pattern Recognition* **10**, 301–312 (1978).
4. L. D. Harmon, M. K. Khan, R. Lasch and P. F. Ramig, Machine identification of human faces, *Pattern Recognition* **13**, 97–110 (1981).
5. C. J. Wu and J. S. Huang, Human face profile recognition by computer, *Pattern Recognition* **23**, 255–259 (1990).

Table 3. Recognition rate by the EBP algorithm and the proposed neuro-fuzzy algorithm as a function of N

Recognition method	N	Recognition rate (%) *db*1	*db*2	*db*3	*db*4	Average
EBP algorithm	10	90.0	92.5	92.5	90.0	91.3
	15	90.0	90.0	91.7	86.7	89.6
	20	90.0	88.8	91.3	82.5	88.2
Neuro-fuzzy algorithm	10	92.5	95.0	97.5	95.0	95.0
	15	93.3	93.3	96.7	93.3	94.2
	20	93.8	95.0	93.8	93.8	94.1

Table 4. Recognition rate by various methods

Recognition method		Recognition rate (%)
Euclidean distance method		82.2
Error backpropagation algorithm		88.2
Fuzzy algorithm	Triangular membership function	81.0
	Trapezoidal membership function	84.1
Neuro-fuzzy algorithm	Triangular membership function	92.9
	Trapezoidal membership function	94.1

6. T. Aibara, K. Ohue and Y. Oshita, Human face profile recognition by a P-Fourier descriptor, *Opt. Eng.* **32**, 861–863 (1993).
7. Y. K. Ham, S. Y. Lee and R.-H. Park, Knowledge-based face recognition using neural networks, *Proc. Int. Joint Conf. Neural Networks* 2125–2128. Nagoya, Japan (1993).
8. M. D. Kelly, Edge detection in pictures by computer using planning, *Machine Intelligence*, **B**. Meltzer and D. Michie (Eds), pp. 397–409. Edinburgh University Press, Edinburgh (1971).
9. I. Craw, H. Ellis and J. R. Lishman, Automatic extraction of face-features, *Pattern Recognition Lett.* **5**, 183–187 (1987).
10. K. Matsuda, T. Kageyama and T. Aibara, Application of the second order statistics for the recognition of human front faces, *Proc. Second Int. Conf. Image Process.* 543–546. Singapore (1992).
11. C.-L. Huang and C.-W. Chen, Human facial feature extraction for face interpretation and recognition, *Pattern Recognition* **25**, 1435–1444 (1992).
12. K. M. Lim and K. W. Oh, A face recognition system using fuzzy logic, *Proc. First Korea-Japan Joint Conf. Comput. Vis.* 506–509. Seoul, Korea (1991).
13. H. Agawa, G. Xu, N. Nagashima and F. Kishino, Image analysis for face modeling and facial image reconstruction, *Proc. SPIE Vis. Commun. Image Process. 90* **1360**, 1184–1197. Lausanne, Switzerland (1990).
14. G. S. Robinson, Edge detection by compass gradient masks, *Comput. Graphics Image Process.* **6**, 492–501 (1977).
15. M. Kato, I. So, Y. Hashimuma, O. Nakamura and T. Minami, Description and synthesis of facial expression based on isodensity maps, *Visual Computing*, T. L. Kunii (Ed.), pp. 36–56. Springer-Verlag, Tokyo (1992).
16. H. Harasaki, M. Yano and T. Nishitani, Background separation/filtering for videophone applications, *Proc. Intl. Conf. Acoust. Speech Sign. Process. 90*, 1981–1984. Albuquerque, New Mexico (1990).
17. C. R. Giardina and E. R. Dougherty, *Morphological Methods in Image and Signal Processing*, pp. 1–35. Prentice-Hall, New Jersey (1988).
18. R. P. Lippmann, An introduction to computing with neural nets, *IEEE Acoust., Speech Sign. Process. Mag.* **4**, 4–22 (1987).
19. L. A. Zadeh, Fuzzy sets, *Fuzzy Set and Application: Selected Papers*, R. R. Yager, S. Ovchinnikov, R. M. Tong and H. T. Nguyen (Eds), pp. 29–44. John Wiley & Sons, New York (1987).
20. G. J. Klir and T. A. Folger, *Fuzzy Sets, Uncertainty, and Information*. Prentice-Hall, New Jersey (1988).
21. Y. K. Ham, S. Y. Lee and R.-H. Park, Fuzzy-based recognition of human front faces using the trapezoidal membership function, *Proc. FUZZ-IEEE/IFES 95* **IV**, 1799–1806. Yokohama, Japan (1995).
22. P. Y. Glorennec, Learning algorithms for neuro-fuzzy networks, *Fuzzy Control Systems*, A. Kandel and G. Langholz (Eds), pp. 3–18. CRC Press, Florida (1994).
23. H. R. Berenji, Fuzzy logic controller, *An Introduction to Fuzzy Logic Applications in Intelligent Systems*, R. R. Yager and L. A. Zadeh (Eds), pp. 69–96. Kluwer Academic Publishers, Massachussetts (1992).

About the Author—SANG YOUNG LEE was born in Buan, Korea, in 1968. He received the B.S. degree in Electronics Engineering from the Korea Institute of Technology, Taejeon, Korea, in 1992 and M.S. degree in Electronics Engineering from Sogang University, Seoul, Korea, in 1994. He has been with the Kia Technical Center, Kia Motors Corporation, since 1994. His research interests are pattern recognition and computer vision.

About the Author—YOUNG K. HAM was born in Kangnung, Korea, in 1966. He received the B.S. and M.S. and Ph.D. degrees in Electronics Engineering from Sogang University, Seoul, Korea, in 1990, 1992 and 1996, respectively. He is currently with the Institute for Defence Information Systems. His research interests are pattern recognition and computer vision.

About the Author—RAE-HONG PARK received the B.S. and M.S. degrees in Electronics Engineering from Seoul National University, Seoul, Korea, in 1976 and 1979, respectively. He received the M.S. and Ph.D. degrees in Electrical Engineering from Stanford University, Stanford, California, in 1981 and 1984, respectively. He joined the faculty of the Department of Electronic Engineering at Sogang University, Seoul, Korea, in 1984, where he is currently a professor. In 1990 he spent his sabbatical year at the Computer Vision Laboratory of the Center for Automation Research, University of Maryland, College Park, Maryland, as a visiting associate professor. His current research interests are computer vision, pattern recognition and video communication. He is a member of the Pattern Recognition Society, KITE, KICS, KISS and IEEE.